

Statistics

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Section 1

Data and frequencies

Overview

Statistics collects, organizes, analyzes and presents data.

Population and sample

We care about a population, but usually observe only a sample.

Descriptive statistics

Descriptive statistics summarize observed data.

Inference

Inference uses sample information to make statements about a population.

Convention. For realized data $x_1, \dots, x_N$, N denotes the number of observations.

Lower-case x_i will denote observed values; later, upper-case X_i will denote random variables.

Types of variables

CATEGORICAL VARIABLES

Values are labels or categories.

Nominal no meaningful order,
e.g. sector or region.

Ordinal a meaningful order, e.g.
low, medium, high.

QUANTITATIVE VARIABLES

Values are numbers.

Discrete a finite or countable set;
counts are a common case.

Continuous values on an interval,
e.g. temperature or duration.

Ordered values: ordinal and quantitative values; nominal labels are not ordered.

Video: [MarinStatsLectures](#), *Variables and types of variables*.

Counting values

For a discrete variable X , let N denote the number of observations.

Frequency	Symbol	Definition
Absolute	$n(x)$	number of observations equal to x
Relative	$r(x)$	$n(x)/N$
For ordered values:		
Cumulative absolute	$N(x)$	$\sum_{y \leq x} n(y)$
Cumulative relative	$R(x)$	$N(x)/N$

Cumulative frequencies require ordered values. They apply to ordinal and quantitative variables.

A small frequency table

Data: 1, 2, 2, 2, 2, 3, 3, 3, 4, 4.

x	1	2	3	4
$n(x)$	1	4	3	2
$r(x)$	.10	.40	.30	.20
$R(x)$	.10	.50	.80	1.00

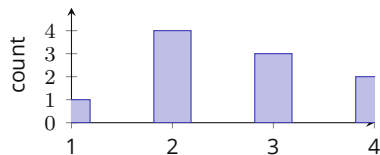


Figure 1 · A bar chart of the data.

Bar heights represent frequencies at the distinct values 1, 2, 3, 4.

Video: [StatQuest, Histograms, Clearly Explained.](#)

Classify the variables

Classify each variable as categorical nominal, categorical ordinal, quantitative discrete, or quantitative continuous:

- (a) region;
- (b) credit rating: low, medium, high;
- (c) number of children;
- (d) annual income;
- (e) waiting time;
- (f) employment status.

Classify the variables

region	categorical nominal
credit rating	categorical ordinal
number of children	quantitative discrete
annual income	quantitative continuous
waiting time	quantitative continuous
employment status	categorical nominal

The order in a credit rating does not make it a numerical measurement; counting children produces a quantitative variable with discrete values.

Section 2

Centre and spread

Mean, median, mode

Mean

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i.$$

Median

the middle value after ordering; for even N , the average of the two middle values.

Mode

the most frequent value or values.

In skewed data, the mean is pulled toward the tail, while the median is less sensitive to extreme observations. On 1, 2, 2, 2, 2, 3, 3, 3, 4, 4 the three summaries are $\bar{x} = 2.6$, median = 2.5 and mode = 2.

Video: [Khan Academy](#), *Central tendency: arithmetic mean, median and mode*.

Quantiles and quartiles

After ordering the observations, quantiles describe positions in the distribution.

Quantile.

A p -quantile q_p is a cutoff with roughly a fraction p of the observations at or below it.

Quartiles.

$$Q_1 = q_{0.25}, \quad Q_2 = q_{0.50}, \quad Q_3 = q_{0.75}.$$

Q_2 is the median.

A sample quantile is obtained from the cumulative relative frequency R ; later, population quantiles are obtained from the CDF F_X in the same way.

Video: [MarinStatsLectures, Percentiles, quantiles and quartiles.](#)

Quartiles by hand: odd and even N

Order the data, cut it at the median, and take the median of each half. With odd N the median is a single observation and is left out of both halves; with even N every observation belongs to one half.

Odd N : nine observations

2, 4, 4, 5, **7**, 8, 9, 12, 15

$Q_2 = 7$ (the fifth value).

Lower half 2, 4, 4, 5: $Q_1 = \frac{4+4}{2} = 4$.

Upper half 8, 9, 12, 15: $Q_3 = \frac{9+12}{2} = 10.5$.

Even N : eight observations

3, 5, 6, **8**, **9**, 11, 14, 20

$Q_2 = \frac{8+9}{2} = 8.5$ (no single middle value).

Lower half 3, 5, 6, 8: $Q_1 = \frac{5+6}{2} = 5.5$.

Upper half 9, 11, 14, 20: $Q_3 = \frac{11+14}{2} = 12.5$.

Software that interpolates between order statistics can report slightly different Q_1 and Q_3 on the same data.

Spread around the centre

Variance

$$s_N^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 \quad \text{average squared deviation}$$

Standard deviation

$$s_N = \sqrt{s_N^2} \quad \text{same units as } X$$

Coefficient of variation

$$CV = \frac{s_N}{\bar{x}} \quad \text{unit-free; for positive ratio-scale variables } (\bar{x} > 0)$$

Here s_N^2 uses denominator N to summarize the observed dataset; the next slide sets out when $N - 1$ is used instead. Other common measures are the range and the average absolute deviation.

Video: [StatQuest, Calculating the mean, variance and standard deviation.](#)

Sample variance for inference

Both denominators divide the same sum of squared deviations $\sum_{i=1}^N (x_i - \bar{x})^2$, and each answers a different question.

Descriptive: divide by N .

$$s_N^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2$$

The empirical distribution $\hat{p}_N$ assigns probability $1/N$ to each observation. Its variance is therefore the average squared deviation in the observed dataset.

Inferential: divide by $N - 1$.

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2, \quad N \geq 2$$

Centring gives $\sum_i (x_i - \bar{x}) = 0$, so only $N - 1$ deviations are free. Under i.i.d. sampling with finite variance, this divisor makes s^2 unbiased for the population variance.

For the same dataset, $s^2 = \frac{N}{N-1} s_N^2$. The descriptive slides that follow report s_N^2 and $s_{xy,N}$; inferential formulas use s^2 and s_{xy} .

Video: [zedstatistics](#), *Variance and standard deviation: why divide by n-1*.

Why the divisor is $N - 1$

Unbiasedness of s^2

Treat the observations as random variables $X_1, \dots, X_N$ drawn independently from the same distribution, with population variance σ^2 . Centring at $\bar{X}$ removes one degree of freedom, and

$$E\left[\sum_{i=1}^N (X_i - \bar{X})^2\right] = (N - 1)\sigma^2.$$

Dividing by $N - 1$ therefore gives $E[s^2] = \sigma^2$, while s_N^2 has expectation $\frac{N-1}{N}\sigma^2$.

Expectation, population variance and i.i.d. sampling are defined in the Probability section.

Worked example

For 1, 2, 2, 2, 2, 3, 3, 3, 4, 4:

CENTRE

Since $\sum_{i=1}^{10} x_i = 26$,

$$\bar{x} = \frac{26}{10} = 2.6.$$

The fifth and sixth ordered values are 2 and 3, so

$$\text{median} = \frac{2 + 3}{2} = 2.5.$$

The value 2 occurs four times, so the mode is 2.

SPREAD

Since $\sum_{i=1}^{10} x_i^2 = 76$,

$$s_N^2 = \frac{76}{10} - \left(\frac{26}{10}\right)^2 = 0.84,$$

$$s_N = \sqrt{0.84} \approx 0.916,$$

$$CV = \frac{s_N}{\bar{x}} \approx \frac{0.916}{2.6} = 0.352,$$

$$s^2 = \frac{10}{9} s_N^2 \approx 0.933.$$

The larger observations at 3 and 4 pull the mean slightly above the median. The mode reports only the most frequent value.

Age and salary

The ten observations are

i	1	2	3	4	5	6	7	8	9	10
Age	18	27	26	18	25	22	27	19	21	23
Salary	750	1000	1750	500	1000	900	1500	950	1300	2500

- Compute the mode of Age.
- Compute the mean, descriptive standard deviation, and coefficient of variation of Salary.
- Group Age into 18–19, 20–21, ..., 26–27 and sketch a histogram.

Age and salary: numerical summaries

AGE

The two modal ages are

18 and 27.

The five adjacent two-year intervals have frequencies

3, 1, 2, 1, 3.

SALARY

$$\sum_{i=1}^{10} y_i = 12150, \quad \sum_{i=1}^{10} y_i^2 = 17777500.$$

Hence

$$\begin{aligned} \bar{y} &= \frac{12150}{10} = 1215, \\ s_N^2 &= \frac{17777500}{10} - 1215^2 = 301525, \\ s_N &\approx 549.11, \quad CV = \frac{s_N}{\bar{y}} \approx 0.452. \end{aligned}$$

Age and salary: grouped ages

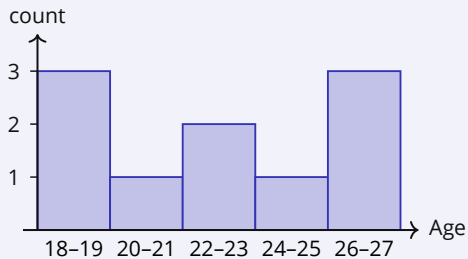


Figure 2 · Age grouped into adjacent two-year bins.

The bars touch because the horizontal axis represents adjacent numerical intervals.

Quartiles and the two variance divisors

Use the quartile convention from this section.

- (a) Find Q_1, Q_2, Q_3 for 1, 2, 4, 5, 7, 8, 9, 12, 15.
- (b) Find Q_1, Q_2, Q_3 for 1, 2, 3, 4, 6, 8, 9, 12.
- (c) For 1, 2, 3, compute s_N^2 and s^2 .

Quartiles and the two variance divisors

For the nine observations, the median is 7; omitting it from the two halves gives

$$(Q_1, Q_2, Q_3) = (3, 7, 10.5).$$

For the eight observations,

$$(Q_1, Q_2, Q_3) = (2.5, 5, 8.5).$$

For 1, 2, 3, $\bar{x} = 2$ and the squared deviations sum to 2, so

$$s_N^2 = \frac{2}{3}, \quad s^2 = \frac{2}{3-1} = 1.$$

Section 3

Shape of a distribution

Skewness & kurtosis

Skewness.

$$\frac{1}{Ns_N^3} \sum_{i=1}^N (x_i - \bar{x})^3$$

Defined when $s_N > 0$.

Positive skewness: greater weight in the right tail.

Negative skewness: greater weight in the left tail.

Video: [ritvikmath, Skewness and kurtosis: two complementary shape summaries.](#)

Kurtosis.

$$\frac{1}{Ns_N^4} \sum_{i=1}^N (x_i - \bar{x})^4$$

Defined when $s_N > 0$.

Normal: kurtosis = 3; sample kurtosis need not be 3.

Excess kurtosis is kurtosis minus 3; normal has 0.

Three shapes by skewness

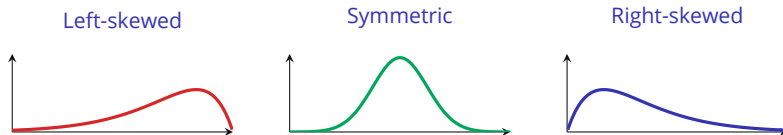


Figure 3 · Left-skewed, symmetric, and right-skewed distribution shapes.

One empirical distribution, many summaries

For realized data $x_1, \dots, x_N$, the relative frequencies define the **empirical distribution**:

$$\hat{p}_N(x) = \frac{n(x)}{N} = r(x), \quad \hat{F}_N(t) = \sum_{x \leq t} \hat{p}_N(x) = R(t).$$

The earlier descriptive quantities are summaries of this same object. For example,

$$\bar{x} = \sum_x x \hat{p}_N(x), \quad s_N^2 = \sum_x (x - \bar{x})^2 \hat{p}_N(x).$$

Quantiles are obtained from $\hat{F}_N$; skewness and kurtosis are standardized third and fourth empirical moments.

Shape, skewness, mean, and median

Consider

$$A = (1, 2, 3, 4, 5), \quad B = (1, 2, 3, 4, 20), \quad C = (-20, -4, -3, -2, -1).$$

- (a) Classify each dataset as symmetric, right skewed, or left skewed.
- (b) Predict the sign of its empirical skewness and compare mean with median.
- (c) Verify directly that the empirical skewness of A is zero.

Shape, skewness, mean, and median

Dataset A is symmetric around 3, so its mean and median are both 3 and its skewness is zero. Indeed, the centered cubes are $(-2)^3, (-1)^3, 0, 1^3, 2^3$, which sum to zero.

Dataset B has a long right tail: its median is 3, its mean is 6, and its skewness is positive.

Dataset $C = -B$ has a long left tail: its median is -3 , its mean is -6 , and its skewness is negative.

Section 4

Joint distributions

Joint, marginal, conditional

For observed pairs (x_i, y_i) , define the empirical distribution.

Joint $\hat{p}_{XY}(x, y) = n_{XY}(x, y) / N.$

Marginal $\hat{p}_X(x) = \sum_y \hat{p}_{XY}(x, y), \quad \hat{p}_Y(y) = \sum_x \hat{p}_{XY}(x, y).$

Conditional $\hat{p}_{X|Y}(x | y) = \hat{p}_{XY}(x, y) / \hat{p}_Y(y), \quad \hat{p}_Y(y) > 0.$

Basic identity

Joint = conditional $\times$ marginal.

$$\hat{p}_{XY}(x, y) = \hat{p}_{X|Y}(x | y) \hat{p}_Y(y), \quad \hat{p}_Y(y) > 0.$$

Video: [zedstatistics, Joint, marginal and conditional probability.](#)

Signal table exercise

JOINT COUNTS

	$s = g$	$s = b$	Total
$\theta = H$	36	14	50
$\theta = L$	18	32	50
Total	54	46	100

CONDITIONAL SHARES

Condition on g

$$\hat{P}(H | g) = \frac{36}{54} = \frac{2}{3}.$$

Condition on b

$$\hat{P}(H | b) = \frac{14}{46}.$$

The good signal realization is more favorable to the hypothesis $\theta = H$.

A joint frequency table

The joint counts are

	$Y = 0$	$Y = 1$	Total
$X = A$	30	10	40
$X = B$	20	40	60
Total	50	50	100

Find the joint and marginal relative frequencies, find $\hat{P}(X = A \mid Y = 1)$, and decide whether X and Y are empirically independent.

A joint frequency table

The joint relative frequencies are

$$\hat{p}(A, 0) = 0.30, \quad \hat{p}(A, 1) = 0.10, \quad \hat{p}(B, 0) = 0.20, \quad \hat{p}(B, 1) = 0.40.$$

The marginals are

$$\hat{p}_X(A) = 0.40, \quad \hat{p}_X(B) = 0.60, \quad \hat{p}_Y(0) = \hat{p}_Y(1) = 0.50.$$

Thus $\hat{P}(X = A \mid Y = 1) = 0.10/0.50 = 0.20$. Independence fails because $\hat{p}(A, 1) = 0.10 \neq (0.40)(0.50) = 0.20$.

Section 5

Covariance and correlation

Measuring association

Descriptive covariance.

$$s_{xy,N} = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})$$

Correlation.

$$r_{xy} = \frac{s_{xy,N}}{s_{x,N} s_{y,N}} \in [-1, 1]$$

Here the covariance uses denominator N . The sample correlation is defined when $s_{x,N} > 0$ and $s_{y,N} > 0$.

Useful covariance identities

$$s_{xx,N} = s_{x,N}^2.$$

$$s_{xy,N} = s_{yx,N}.$$

$$s_{\alpha x, \beta y, N} = \alpha \beta s_{xy, N}.$$

Video: [DataMListic, Covariance and correlation explained.](#)

The two covariance divisors

Empirical distribution.

$$s_{xy,N} = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})$$

This is the covariance of $\hat{p}_N$, which puts mass $1/N$ on each observed pair (x_i, y_i) .

Estimating population covariance.

$$s_{xy} = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})$$

$$s_{xy} = \frac{N}{N-1} s_{xy,N}$$

The $1/(N-1)$ divisor is the inferential convention, matching s^2 for the variance.

Why the divisor is $N - 1$

Unbiasedness of s_{xy}

For pairs (X_i, Y_i) drawn independently from the same distribution with finite second moments,

$$E[s_{xy}] = \text{Cov}(X, Y), \quad s_{xy} = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y}),$$

by the same degrees-of-freedom count that gives $E[s^2] = \sigma^2$.

Population covariance is defined in the Probability section.

The correlation is unchanged

Each standard deviation changes by the square root of the covariance factor:

$$s_x = \sqrt{\frac{N}{N-1}} s_{x,N}, \quad s_y = \sqrt{\frac{N}{N-1}} s_{y,N},$$

Hence their product has the covariance factor:

$$s_x s_y = \frac{N}{N-1} s_{x,N} s_{y,N}.$$

The common factor cancels:

$$\frac{s_{xy}}{s_x s_y} = \frac{s_{xy,N}}{s_{x,N} s_{y,N}} = r_{xy}.$$

Thus a consistently calculated Pearson sample correlation has the same numerical value under the (N) and (N-1) conventions.

Mean vector and covariance matrix

The three observations of (X_1, X_2, X_3) are

	X_1	X_2	X_3
1	4	5	3
2	7	3	9
3	1	8	6

- (a) Compute the mean vector $\bar{X}$.
- (b) Using divisor $N = 3$, compute the empirical covariance matrix S_N .
- (c) Using divisor $N - 1 = 2$, compute the unbiased covariance estimator $\hat{\Sigma}$.

Video: [ritvikmath, The covariance matrix.](#)

Mean vector

Average each coordinate across the three observations:

$$\bar{X}_j = \frac{1}{3} \sum_{i=1}^3 x_{ij},$$

$$\bar{X}_1 = \frac{4 + 7 + 1}{3} = 4, \quad \bar{X}_2 = \frac{5 + 3 + 8}{3} = \frac{16}{3}, \quad \bar{X}_3 = \frac{3 + 9 + 6}{3} = 6,$$

so

$$\bar{X} = \begin{bmatrix} 4 \\ 16/3 \\ 6 \end{bmatrix}.$$

Only $\bar{X}_2$ fails to come out whole, which is why the centered observations and the covariance matrix below carry thirds.

Center the observations

Subtract $\bar{X} = (4, 16/3, 6)^T$ from each data row:

$$(4, 5, 3) - \bar{X} = (0, -1/3, -3),$$

$$(7, 3, 9) - \bar{X} = (3, -7/3, 3),$$

$$(1, 8, 6) - \bar{X} = (-3, 8/3, 0).$$

Stack those centered rows:

$$D = \begin{bmatrix} 0 & -1/3 & -3 \\ 3 & -7/3 & 3 \\ -3 & 8/3 & 0 \end{bmatrix}.$$

Empirical covariance matrix

With the descriptive $1/N$ convention,

$$S_N = \frac{1}{3} D^T D = \begin{bmatrix} 6 & -5 & 3 \\ -5 & 38/9 & -2 \\ 3 & -2 & 6 \end{bmatrix}.$$

Unbiased covariance estimator

Since $N = 3$, changing the divisor from N to $N - 1$ multiplies every entry of the covariance matrix by $3/2$:

$$\hat{\Sigma} = \frac{1}{2} D^T D = \frac{3}{2} S_N = \begin{bmatrix} 9 & -15/2 & 9/2 \\ -15/2 & 19/3 & -3 \\ 9/2 & -3 & 9 \end{bmatrix}.$$

Video: [StatQuest, Covariance, Clearly Explained](#).

Association and causation

A positive sample correlation,

$$r_{\text{education, wage}} > 0,$$

shows association in the data.

A causal claim additionally needs a design that isolates the channel of interest.

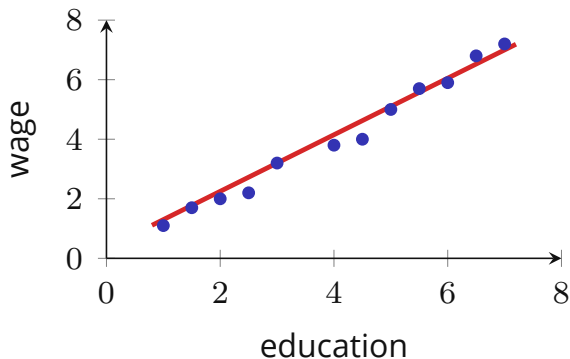


Figure 4 · Education and wages.

Association alone does not identify causation.

Video: *Sprouts, Correlation versus causation.*

Direct and omitted channels

Direct channel. More schooling may raise productivity and wages.

Omitted channel. Ability may raise both schooling and wages.

Statistical and causal objects

Statistical object: association, which the covariance and the correlation measure.

Causal object: the effect of schooling on wages, which requires identifying variation, such as a design, an instrument, an experiment, or restrictions imposed by a model.

Compute and transform a correlation

For the three observed pairs

$$(X, Y) = (1, 1), (2, 3), (3, 2),$$

compute $s_{xy,N}$, $s_{x,N}$, $s_{y,N}$, and r_{xy} . Then replace Y by $Z = 2Y + 5$ and verify that $r_{xz} = r_{xy}$.

Compute and transform a correlation

Here $\bar{x} = \bar{y} = 2$, with centered vectors $(-1, 0, 1)$ and $(-1, 1, 0)$. Hence

$$s_{xy,N} = \frac{1}{3}, \quad s_{x,N} = s_{y,N} = \sqrt{\frac{2}{3}}, \quad r_{xy} = \frac{1/3}{2/3} = \frac{1}{2}.$$

For $Z = 2Y + 5$, $s_{xz,N} = 2/3$ and $s_{z,N} = 2\sqrt{2/3}$. Therefore

$$r_{xz} = \frac{2/3}{\sqrt{2/3} 2\sqrt{2/3}} = \frac{1}{2} = r_{xy}.$$

Section 6

Probability & random variables

From observed data to a probability model

The first half of the lecture described realized data. Probability now introduces a model for the process that can generate such data.

Observed data.

Observed values $x_1, \dots, x_N$ determine

$$\hat{p}_N, \quad \hat{F}_N, \quad \bar{x}, \quad s_N^2.$$

These are fixed once the data are observed.

Population.

A random variable X has a population distribution with objects such as

$$p_X \text{ or } f_X, \quad F_X, \quad E[X], \quad \text{Var}(X).$$

These describe the probabilistic model.

Statistical inference studies what the observed quantities reveal about the corresponding population quantities.

Sample space, events, and probability

A probability model begins with possible outcomes and events constructed from them.

Sample space Ω the set of all possible outcomes ω .

Event A a collection of outcomes; formally, $A \in \mathcal{F}$.

Probability P assigns $P(A) \in [0, 1]$ to events, with $P(\Omega) = 1$.

For events A and B , two basic identities are

$$P(A^c) = 1 - P(A),$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Here $\mathcal{F}$ is the **collection of events** to which probabilities can be assigned.

Video: [Steve Brunton, Set theory in probability: sample spaces and events.](#)

From outcomes to numbers

On a probability space $(\Omega, \mathcal{F}, P)$, a real-valued random variable $X: \Omega \rightarrow \mathbb{R}$ assigns a number to each outcome. Events about X are preimages:

$$\{X \leq t\} = X^{-1}((-\infty, t]) = \{\omega \in \Omega : X(\omega) \leq t\}.$$

Formal wording

Formally X is measurable, $X: (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, meaning $X^{-1}(B) \in \mathcal{F}$ for every Borel set $B \in \mathcal{B}(\mathbb{R})$, so probabilities of restrictions on X are well defined.

Number of heads

Toss a fair coin twice and let X be the number of heads. Then

ω	HH	HT	TH	TT
$X(\omega)$	2	1	1	0

so the induced distribution is

$$\Pr(X = 0) = \frac{1}{4}, \quad \Pr(X = 1) = \frac{1}{2}, \quad \Pr(X = 2) = \frac{1}{4}.$$

An event stated in terms of X is a set of underlying outcomes:

$$\{X \geq 1\} = \{HH, HT, TH\}, \quad \Pr(X \geq 1) = \frac{3}{4}.$$

Discrete and continuous distributions

For either type of distribution, the cumulative distribution function is $F_X(t) = \Pr(X \leq t)$.

Discrete.

A probability mass function p satisfies

$$p(x) = \Pr(X = x), \quad p(x) \geq 0, \\ \sum_x p(x) = 1.$$

CDF

$$F_X(t) = \sum_{x \leq t} p(x).$$

Density.

For every Borel set A , a density f satisfies

$$\Pr(X \in A) = \int_A f(x) dx, \quad f(x) \geq 0, \\ \int_{\mathbb{R}} f(x) dx = 1.$$

CDF

$$F_X(t) = \int_{-\infty}^t f(x) dx.$$

Video: [RiskByNumbers](#), *Probability distributions explained visually*.

The Bernoulli and the normal

BERNOULLI

If $X \sim \text{Bernoulli}(p)$, then

$$\Pr(X = x) = \begin{cases} 1 - p, & x = 0, \\ p, & x = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Hence

$$F_X(t) = \begin{cases} 0, & t < 0, \\ 1 - p, & 0 \leq t < 1, \\ 1, & t \geq 1. \end{cases}$$

UNIFORM ON $[0, 1]$

If $Y \sim \text{Uniform}(0, 1)$, then

$$f_Y(y) = \begin{cases} 1, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Hence

$$F_Y(t) = \begin{cases} 0, & t < 0, \\ t, & 0 \leq t \leq 1, \\ 1, & t > 1. \end{cases}$$

Thus $\Pr(0.2 \leq Y \leq 0.7) = 0.5$.

Expectation and variance

Discrete.

$$E[X] = \sum_x x p(x).$$

$$\text{Var}(X) = E[(X - E[X])^2].$$

Continuous.

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx.$$

$$\text{Var}(X) = E[X^2] - E[X]^2.$$

The second identity holds in both cases, since $E[(X - E[X])^2] = E[X^2] - E[X]^2$ whenever the second moment is finite.

The expectation formulas require integrability and the variance formulas require a finite second moment. These are **population** moments. (In comparison, the **sample** moments from the first half of the lecture estimate them from **data**.)

Video: [Dr. Gajendra Purohit, Mathematical expectation and variance of discrete and continuous random variables.](#)

Bernoulli mean and variance

Let $X \sim \text{Bernoulli}(p)$. Since $X \in \{0, 1\}$,

$$E[X] = 0 \cdot (1 - p) + 1 \cdot p = p.$$

Also $X^2 = X$, so

$$E[X^2] = p.$$

Therefore

$$\text{Var}(X) = E[X^2] - E[X]^2 = p - p^2 = p(1 - p).$$

For a fair coin indicator, $p = \frac{1}{2}$, so $E[X] = \frac{1}{2}$ and $\text{Var}(X) = \frac{1}{4}$.

A discrete law and a continuous density

- (a) Let $P(X = 0) = 0.2$, $P(X = 1) = 0.5$, and $P(X = 2) = 0.3$. Find $F_X(t)$, $P(X \geq 1)$, $E[X]$, and $Var(X)$.
- (b) Let Y have density $f(y) = 2y$ on $[0, 1]$, and zero elsewhere. Verify that this is a density, find $F_Y(y)$, and compute $P(Y > 1/2)$ and $E[Y]$.

The discrete distribution

$$F_X(t) = \begin{cases} 0, & t < 0, \\ 0.2, & 0 \leq t < 1, \\ 0.7, & 1 \leq t < 2, \\ 1, & t \geq 2. \end{cases}$$

Also,

$$P(X \geq 1) = 0.8, \quad E[X] = 1.1, \quad E[X^2] = 1.7,$$

$$\text{so } \text{Var}(X) = 1.7 - (1.1)^2 = 0.49.$$

The continuous distribution

Since $2y \geq 0$ on $[0, 1]$ and $\int_0^1 2y \, dy = 1$, the function is a density. Its CDF is

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ y^2, & 0 \leq y \leq 1, \\ 1, & y > 1. \end{cases}$$

Therefore

$$P(Y > 1/2) = 1 - F_Y(1/2) = \frac{3}{4}, \quad E[Y] = \int_0^1 y(2y) \, dy = \frac{2}{3}.$$

Population objects

The descriptive covariance introduced earlier averages products over the observed sample; population covariance uses expectation under the probability distribution instead.

Covariance.

For random variables X and Y with finite second moments,

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY] - E[X]E[Y].\end{aligned}$$

Independence.

The joint distribution factors into the marginal distributions. In the discrete case,

$$p_{XY}(x, y) = p_X(x)p_Y(y) \quad \text{for all } x, y.$$

With a joint density,

$$f_{XY}(x, y) = f_X(x)f_Y(y).$$

Independence implies $\text{Cov}(X, Y) = 0$ whenever the covariance exists; the converse generally fails. If both variances are positive, $\text{Corr}(X, Y) = \text{Cov}(X, Y) / \sqrt{\text{Var}(X)\text{Var}(Y)}$, the population analogue of the descriptive correlation defined earlier.

Zero covariance need not mean independence

Let

$$\Pr(X = -1) = \frac{1}{4}, \quad \Pr(X = 0) = \frac{1}{2}, \quad \Pr(X = 1) = \frac{1}{4}, \quad Y = X^2.$$

Symmetry gives $E[X] = 0$ and $E[XY] = E[X^3] = 0$, so

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0.$$

Nevertheless X and Y fail the factorization test for independence:

$$\Pr(X = 0, Y = 0) = \frac{1}{2}, \quad \Pr(X = 0) \Pr(Y = 0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Covariance sees linear co-movement, so it can miss a deterministic nonlinear relation. The other direction does hold: independence forces zero covariance when the second moments exist, so a nonzero covariance proves dependence. For jointly normal (X, Y) , zero covariance gives independence.

Independence and covariance: logical directions

Assume the required second moments exist. Decide whether each statement is true or false.

- (a) Independence implies zero covariance.
- (b) Zero covariance implies independence.
- (c) Nonzero covariance implies dependence.
- (d) For jointly normal variables, zero covariance implies independence.

Independence and covariance: logical directions

- (a) True: factorization makes $E[XY] = E[X]E[Y]$.
- (b) False: $Y = X^2$ in the preceding example is a counterexample.
- (c) True: it is the contrapositive of independence implying zero covariance.
- (d) True: within a jointly normal distribution, zero covariance makes the joint Gaussian law factorize into its marginal laws.

Linear combinations & covariance matrices

Two variables.

Let $Z = \alpha_1 X_1 + \alpha_2 X_2$ with finite second moments.

$$E[Z] = \alpha_1 E[X_1] + \alpha_2 E[X_2].$$

$$\begin{aligned} \text{Var}(Z) &= \alpha_1^2 \text{Var}(X_1) + \alpha_2^2 \text{Var}(X_2) \\ &\quad + 2\alpha_1 \alpha_2 \text{Cov}(X_1, X_2). \end{aligned}$$

Matrix form.

In k dimensions, let $\Sigma = [\text{Cov}(X_i, X_j)]$.

For every $\alpha \in \mathbb{R}^k$,

$$\text{Var}(\alpha'X) = \alpha' \Sigma \alpha \geq 0.$$

Hence every covariance matrix is positive semidefinite.

Both statements are the population versions of the descriptive covariance identities: scaling a variable scales the covariance, and a variance is a covariance of a variable with itself.

Sample quantities and population objects

Descriptive statistics are computed from realized data; population objects describe the probability distribution used to model the data.

Sample / empirical object	Population object	Role
$r(x)$	$p_X(x)$	relative frequency / probability mass
$R(x)$	$F_X(x)$	cumulative frequency / CDF
$\bar{x}$	$E[X]$	centre
s_N^2	$Var(X)$	spread
$s_{xy,N}$	$Cov(X, Y)$	linear co-movement

Statistical inference studies how sample objects can be used to learn about population objects.

Section 7

Conditioning & Bayes

Conditioning, independence, and total probability

Conditional probability.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

Independence.

$$P(A \cap B) = P(A)P(B).$$

Equivalently, $P(A \mid B) = P(A)$ whenever the conditional probability is defined.

Total probability

If $A_1, \dots, A_m$ partition the state space, then

$$P(B) = \sum_{j=1}^m P(B \mid A_j)P(A_j).$$

Video: [jbstatistics](#), *The law of total probability*.

A conditional average

Conditional expectation and iterated expectations

Once a conditional distribution is known, it can be summarized by its conditional expectation. For discrete Y ,

$$E[Y | X = x] = \sum_y y p_{Y|X}(y | x).$$

AT A REALIZED VALUE

For fixed x , $E[Y | X = x]$ is a number: the mean conditional on $X = x$.

AS A RANDOM VARIABLE

$E[Y | X]$ is a function of X , and therefore is itself a random variable.

Law of iterated expectations

$$E[E[Y | X]] = E[Y].$$

Further reading: MIT 18.05, [probability and statistics readings](#).

Conditional expectation and iterated expectations

Let $X, Y \in \{0, 1\}$ have joint probabilities

	$Y = 0$	$Y = 1$	$P(X = x)$
$X = 0$	0.2	0.3	0.5
$X = 1$	0.1	0.4	0.5

Compute $E[Y | X = 0]$, $E[Y | X = 1]$, and $E[Y]$. Then verify $E[E[Y | X]] = E[Y]$.

Conditional expectation and iterated expectations

$$E[Y | X = 0] = \frac{0.3}{0.5} = 0.6, \quad E[Y | X = 1] = \frac{0.4}{0.5} = 0.8.$$

Since $P(Y = 1) = 0.3 + 0.4 = 0.7$, $E[Y] = 0.7$. Averaging the two conditional means over X gives

$$E[E[Y | X]] = (0.6)(0.5) + (0.8)(0.5) = 0.7 = E[Y].$$

Updating from a signal

Writing the joint probability of A and B in the two possible orders,

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A),$$

and dividing by $P(B)$ gives

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}, \quad P(B) > 0.$$

Here A is a state of the world and B an observed signal realization. The prior $P(A)$ is reweighted by the likelihood $P(B \mid A)$ of that realization in the state, and the denominator $P(B) = \sum_j P(B \mid A_j)P(A_j)$ comes from the law of total probability over a partition of the states.

Video: [3Blue1Brown](#), *The medical test paradox and Bayes' rule*.

A signal update

The state of the economy is unobserved: Boom or Bust. An analyst observes a noisy indicator, High or Low. A High outcome is more likely in a boom.

Prior and signal model

$$P(\text{Boom}) = P(\text{Bust}) = \frac{1}{2},$$

$$P(\text{High} \mid \text{Boom}) = 0.8,$$

$$P(\text{High} \mid \text{Bust}) = 0.3.$$

After observing High

$$P(\text{High}) = 0.8(0.5) + 0.3(0.5) = 0.55,$$

$$P(\text{Boom} \mid \text{High}) = \frac{0.8(0.5)}{0.55} \approx 0.727.$$

A High outcome does not reveal the state. It raises the probability of a boom from 0.5 to 0.727.

The average posterior equals the prior

Bayes plausibility

Let $\theta \in \{0, 1\}$, with prior $\mu_0 = P(\theta = 1)$. After signal S , the posterior is $\mu(S) = P(\theta = 1 \mid S)$. Iterated expectations give

$$E[\mu(S)] = E[P(\theta = 1 \mid S)] = P(\theta = 1) = \mu_0.$$

Thus every feasible distribution τ of binary-state posteriors satisfies

$$\int_0^1 \mu \, d\tau(\mu) = \mu_0.$$

Every Bayes-plausible distribution can be implemented

A direct construction

Let τ be any distribution on $[0, 1]$ satisfying $\int_0^1 m d\tau(m) = \mu_0$. Define the joint distribution by

$$P(\theta = 1, \mu \in dm) = m d\tau(m), \quad P(\theta = 0, \mu \in dm) = (1 - m) d\tau(m).$$

Its marginal distribution of μ is τ , while

$$P(\theta = 1) = \int_0^1 m d\tau(m) = \mu_0, \quad P(\theta = 1 \mid \mu = m) = m$$

for τ -almost every m . Thus the mean condition is also sufficient.

A repayment signal

A borrower repays with probability $P(R) = 0.6$. A screening signal takes its favourable realization G with

$$P(G | R) = 0.8, \quad P(G | R^c) = 0.3.$$

- (a) Write the joint distribution of R and G .
- (b) Compute $P(G)$ and $P(R | G)$.
- (c) Compute $E[\mathbf{1}_R]$, $Var(\mathbf{1}_R)$, and $E[\mathbf{1}_R | \mathbf{1}_G = 1]$.
- (d) Compute $Cov(\mathbf{1}_R, \mathbf{1}_G)$ and decide whether R and G are independent.

A repayment signal

The joint probabilities are

	G	G^c	Total
R	0.48	0.12	0.60
R^c	0.12	0.28	0.40
Total	0.60	0.40	1

because $P(R \cap G) = 0.6(0.8) = 0.48$ and $P(R^c \cap G) = 0.4(0.3) = 0.12$. Therefore

$$P(G) = 0.60, \quad P(R | G) = \frac{0.48}{0.60} = 0.8.$$

A repayment signal

The indicator $\mathbf{1}_R$ is Bernoulli with success probability $P(R) = 0.6$, so

$$E[\mathbf{1}_R] = P(R) = 0.6, \quad \text{Var}(\mathbf{1}_R) = 0.6(0.4) = 0.24, \quad E[\mathbf{1}_R \mid \mathbf{1}_G = 1] = P(R \mid G) = 0.8.$$

Finally $E[\mathbf{1}_R \mathbf{1}_G] = P(R \cap G) = 0.48$, so

$$\text{Cov}(\mathbf{1}_R, \mathbf{1}_G) = 0.48 - (0.6)(0.6) = 0.12.$$

The covariance is nonzero, so R and G are not independent.

From a random sample to a sampling distribution

Let $X_1, \dots, X_N$ be a random sample and let $x_1, \dots, x_N$ denote one realized dataset.

Statistic. A statistic is a function of the random sample,

$$T_N = T(X_1, \dots, X_N).$$

Before observing the data, T_N is a random variable; after realization it takes the value $t_N = T(x_1, \dots, x_N)$.

For example,

$$\bar{X}_N = \frac{1}{N} \sum_{i=1}^N X_i \quad \mapsto \quad \bar{x} = \frac{1}{N} \sum_{i=1}^N x_i.$$

An **estimator** is a statistic used to learn about an unknown population quantity. Its probability distribution across possible samples is its **sampling distribution**.

For the sample mean below, the law of large numbers describes concentration around the population mean; the central limit theorem describes the large-sample shape and scale of the sampling error.

Statistic, estimator, estimate, or parameter?

Classify each object and explain whether it is random before the sample is observed:

$$\bar{X}_N, \quad \bar{x} = 2.7, \quad s^2, \quad \sigma^2.$$

Which objects can be used as estimators, and which object is an estimate?

Statistic, estimator, estimate, or parameter?

$\bar{X}_N$ is a statistic and an estimator of the population mean; it is random before the data are observed. The realized number $\bar{x} = 2.7$ is an estimate. The sample variance s^2 is a statistic and can be used as an estimator of σ^2 . The quantity σ^2 is a population parameter, not a statistic.

Law of large numbers

Large-sample mean

Large-sample mean.

$X_1, \dots, X_N$ are i.i.d. with $E[X_i] = \mu$,
 $E[|X_i|] < \infty$.

$$\bar{X}_N \xrightarrow{p} \mu$$

The sample mean becomes close to the population mean.

Video: MIT 6.041, *The weak law of large numbers*. Further reading: Pishro-Nik, *Introduction to Probability*, §7.1.1.

Central limit theorem

Central limit theorem

Central limit theorem.

$X_1, \dots, X_N$ are i.i.d. with $E[X_i] = \mu$,
 $\text{Var}(X_i) = \sigma^2 \in (0, \infty)$.

$$\frac{\sqrt{N}(\bar{X}_N - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1)$$

The standardized sampling error is approximately normal for large N .

Video: [3Blue1Brown](#), *What Is the Central Limit Theorem?*.

$\xrightarrow{p}$: convergence in probability $\xrightarrow{d}$: convergence in distribution.

Further reading: [MIT 18.05](#), Central Limit Theorem and the Law of Large Numbers.

A sample proportion

A sample proportion

Bernoulli sample

Let $X_1, \dots, X_N$ be i.i.d. $\text{Bernoulli}(p)$, and define

$$\hat{p}_N = \frac{1}{N} \sum_{i=1}^N X_i.$$

$$E[X_i] = p, \quad \text{Var}(X_i) = p(1 - p).$$

Large-sample result

$$\begin{aligned} \hat{p}_N &\xrightarrow{p} p \\ \frac{\hat{p}_N - p}{\sqrt{p(1-p)/N}} &\xrightarrow{d} \mathcal{N}(0, 1). \end{aligned}$$

Numerical scale: $p = 0.5$, $N = 400$

$$\text{se}(\hat{p}_N) = \sqrt{\frac{0.5(1-0.5)}{400}} = 0.025.$$

$$0.5 \pm 1.96(0.025) = [0.451, 0.549].$$

Under the normal approximation, this interval contains $\hat{p}_N$ with probability about 0.95; its half-width is about five percentage points.

Use the LLN and CLT

- (a) Suppose $X_1, \dots, X_N$ are i.i.d., $E[X_i] = 10$, $\text{Var}(X_i) = 16$, and $N = 100$. State the LLN conclusion, give the normal approximation to the distribution of $\bar{X}_N$, and approximate $P(9.2 \leq \bar{X}_N \leq 10.8)$.
- (b) For a Bernoulli sample with $p = 0.4$ and $N = 400$, find $\text{se}(\hat{p}_N)$ and an approximate central 95% range for $\hat{p}_N$.

Use the LLN and CLT

The LLN gives $\bar{X}_N \xrightarrow{p} 10$. For $N = 100$,

$$\bar{X}_N \approx \mathcal{N}\left(10, \frac{16}{100}\right), \quad \text{se}(\bar{X}_N) = 0.4.$$

The two endpoints have standardized values -2 and 2 , so

$$P(9.2 \leq \bar{X}_N \leq 10.8) \approx \Phi(2) - \Phi(-2) \approx 0.9545.$$

For the Bernoulli sample,

$$\text{se}(\hat{p}_N) = \sqrt{\frac{0.4(0.6)}{400}} \approx 0.0245,$$

giving $0.4 \pm 1.96(0.0245) \approx [0.352, 0.448]$.

Section 8

Statistical applications

Exogeneity identifies the OLS coefficient

From conditional mean zero to population identification

In the linear model $Y = X^\top \beta + u$, suppose

$$E[u \mid X] = 0.$$

Then $E[Y \mid X] = X^\top \beta$ and iterated expectations imply

$$E[Xu] = E[XE[u \mid X]] = 0.$$

If $E[XX^\top]$ is nonsingular, the orthogonality condition has the unique solution

$$\beta = E[XX^\top]^{-1}E[XY].$$

Video: [Colleen O'Briant, Assumption of exogeneity](#).

Sample moments connect OLS consistency to IV

Consistency and instrumental variables

Under $E[Xu] = 0$, independent sampling, finite second moments, and nonsingular $E[XX^\top]$,

$$\hat{\beta} - \beta = \left(\frac{1}{N} \sum_{i=1}^N X_i X_i^\top \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N X_i u_i \right) \xrightarrow{p} 0.$$

If $E[Xu] \neq 0$, OLS does not consistently estimate β . An instrument Z replaces the OLS moment by

$$E[Zu] = 0, \quad \text{rank } E[ZX^\top] = \dim(\beta).$$

In the scalar model with an intercept, relevance is $\text{Cov}(Z, X) \neq 0$.

What the OLS and IV conditions do

In the linear model $Y = X^T \beta + u$, match each condition to its role and state what it permits us to conclude:

- (a) $E[u | X] = 0$;
- (b) $E[Zu] = 0$;
- (c) $\text{rank } E[ZX^T] = \dim(\beta)$.

In the scalar model with an intercept, how can condition (c) be written using covariance?

What the OLS and IV conditions do

- (a) $E[u | X] = 0$ is OLS exogeneity. It implies $E[Xu] = 0$, which identifies β when $E[XX^T]$ is nonsingular.
- (b) $E[Zu] = 0$ is instrument exogeneity: the instrument is orthogonal to the structural error.
- (c) The rank condition is instrument relevance. Together with the IV moment, it identifies β . In the scalar model with an intercept, it becomes $\text{Cov}(Z, X) \neq 0$.

OLS under normal errors

This is the same statistic-sampling-distribution logic in vector form. An estimator is a function of random data; its realized numerical value is an estimate.

Linear model

$$y = X\beta + u, \quad u | X \sim \mathcal{N}(0, \sigma^2 I).$$

Conditional on X , the randomness in $\hat{\beta}$ comes from the error vector u .

OLS estimator

If $X'X$ is invertible,

$$\hat{\beta} = \beta + (X'X)^{-1}X'u.$$

Video: [Dr. Venoo Kakar, Sampling distribution of the OLS estimators.](#)

OLS sampling distribution

Under the normal-error model, with $X'X$ invertible and $N > k$, conditional on X :

Conditional distribution

$$\hat{\beta} \mid X \sim \mathcal{N}(\beta, \sigma^2(X'X)^{-1}).$$

Variance estimator

$$\hat{\sigma}^2 = \frac{\|y - X\hat{\beta}\|^2}{N - k}.$$

Standard error

$$\text{se}(\hat{\beta}_j) = \hat{\sigma} \sqrt{[(X'X)^{-1}]_{jj}}.$$

Studentized coefficient

$$\frac{\hat{\beta}_j - \beta_j}{\text{se}(\hat{\beta}_j)} \sim t_{N-k}.$$

Video: Ben Lambert, *Normally distributed errors and finite-sample inference*.